

Homework 6

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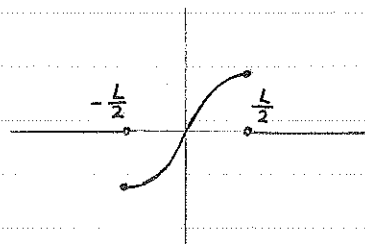
5.1.4

The answer is already given in Shankar.

Moreover, it is important to realize that this cannot be a solution of Schrödinger Equation. Solutions of Schrödinger Equation should satisfy the following:

- ① the wavefunction is continuous;
- ② the first order derivative of the wavefunction is usually continuous for finite potentials.

Therefore, the given wavefunction is not physical (not continuous at $x = \pm \frac{L}{2}$).



5.2.1

Initial state: $\psi_i(x) = \sqrt{\frac{2}{L}} \cos\left(\frac{\pi x}{L}\right) \quad \left(-\frac{L}{2} \leq x \leq \frac{L}{2}\right)$

It is undisturbed under a sudden expansion of the box from L to $2L$.

New ground state: $\phi_1(x) = \sqrt{\frac{1}{L}} \cos\left(\frac{\pi x}{2L}\right) \quad (-L \leq x \leq L)$

The probability of finding the particle in $\phi_1(x)$ is

$$P = |\langle \phi_1 | \psi_i \rangle|^2$$

$$\begin{aligned} \langle \phi_1 | \psi_i \rangle &= \int_{-\infty}^{\infty} \phi_1^*(x) \psi_i(x) dx \\ &= \int_{-\frac{L}{2}}^{\frac{L}{2}} \sqrt{\frac{1}{L}} \cdot \sqrt{\frac{2}{L}} \cos\left(\frac{\pi x}{2L}\right) \cos\left(\frac{\pi x}{L}\right) dx \\ &= \frac{8}{3\pi} \end{aligned}$$

Therefore, $P = \left(\frac{8}{3\pi}\right)^2$.

5.2.2

(a) $|\psi\rangle = \sum_i \alpha_i |\phi_i\rangle$, where ϕ_i is an eigenstate of H with eigenvalue E_i .

$$\begin{aligned} \langle \psi | H | \psi \rangle &= \sum_i \sum_j \alpha_i^* \alpha_j \langle \phi_i^* | H | \phi_j \rangle = \sum_i \sum_j \alpha_i^* \alpha_j E_j \langle \phi_i^* | \phi_j \rangle \\ &= \sum_i \sum_j \alpha_i^* \alpha_j E_j \delta_{ij} = \sum_i |\alpha_i|^2 E_i \geq E_0 \sum_i |\alpha_i|^2 = E_0 \end{aligned}$$

Therefore, $\langle \psi | H | \psi \rangle \geq E_0$.

(b) Consider $\psi_\alpha(x) = \left(\frac{\alpha}{\pi}\right)^{1/4} e^{-\alpha x^2/2}$

$$\frac{d}{dx} \psi_\alpha(x) = -\frac{\alpha}{2} \cdot 2x \left(\frac{\alpha}{\pi}\right)^{1/4} e^{-\alpha x^2/2} = -\alpha x \left(\frac{\alpha}{\pi}\right)^{1/4} e^{-\alpha x^2/2}$$

$$\frac{d^2}{dx^2} \psi_\alpha(x) = -\alpha \left(\frac{\alpha}{\pi}\right)^{1/4} e^{-\alpha x^2/2} + \alpha^2 x^2 \left(\frac{\alpha}{\pi}\right)^{1/4} e^{-\alpha x^2/2}$$

$$\begin{aligned} \langle E(\alpha) \rangle &= \langle \psi_\alpha(x) | -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} - |V(x)| | \psi_\alpha(x) \rangle \\ &= \langle \psi_\alpha(x) | -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} | \psi_\alpha(x) \rangle - \langle \psi_\alpha(x) | |V(x)| | \psi_\alpha(x) \rangle \\ &= \frac{\hbar^2}{2m} \left(\frac{\alpha}{\pi}\right)^{1/2} \left(\int_{-\infty}^{\infty} \alpha e^{-\alpha x^2} dx - \int_{-\infty}^{\infty} \alpha^2 x^2 e^{-\alpha x^2} dx \right) - \int_{-\infty}^{\infty} |V(x)| \left(\frac{\alpha}{\pi}\right)^{1/2} e^{-\alpha x^2} dx \\ &= \frac{\hbar^2}{2m} \left(\frac{\alpha}{\pi}\right)^{1/2} \left(\frac{\alpha}{2} - \int_{-\infty}^{\infty} |V(x)| \left(\frac{\alpha}{\pi}\right)^{1/2} e^{-\alpha x^2} dx \right) \end{aligned}$$

Try to find the minimum value for $\langle E(\alpha) \rangle$.

$$\frac{d}{d\alpha} \langle E(\alpha) \rangle = 0$$

$$\Rightarrow \frac{\hbar^2}{4m} - \left(\frac{\alpha}{\pi}\right)^{1/2} \int_{-\infty}^{\infty} |V(x)| \left(\frac{1}{2\alpha} - x^2\right) e^{-\alpha x^2} dx = 0$$

$$\Rightarrow \alpha = \frac{\int_{-\infty}^{\infty} |V(x)| \psi_\alpha^2(x) dx}{\frac{\hbar^2}{2m} + 2 \int_{-\infty}^{\infty} |V(x)| x^2 \psi_\alpha^2(x) dx}$$

$$\langle E(\alpha) \rangle = \frac{\int_{-\infty}^{\infty} |V(x)| \psi_\alpha^2(x) dx}{2 + \frac{2m}{\hbar^2} \int_{-\infty}^{\infty} |V(x)| x^2 \psi_\alpha^2(x) dx} - \int_{-\infty}^{\infty} |V(x)| \psi_\alpha^2(x) dx < 0$$

According to part (a), $E_0 \leq \langle E(\alpha) \rangle < 0$. Therefore, there exists at least a bound state with $E < 0$.

5.2.3

* Boundary condition for δ -potential.

$$\int_{-\epsilon}^{\epsilon} \psi'' dx = \psi'(x) \Big|_{-\epsilon}^{\epsilon} = \psi'_I(\epsilon) - \psi'_I(-\epsilon) \quad (1)$$

Meanwhile,

$$\text{since } \psi'' + \frac{2m}{\hbar^2} [E + V_0 a \delta(x)] \psi = 0$$

$$\begin{aligned} \int_{-\epsilon}^{\epsilon} \psi'' dx &= - \int_{-\epsilon}^{\epsilon} \frac{2m}{\hbar^2} [E + V_0 a \delta(x)] \psi dx \\ &= - \int_{-\epsilon}^{\epsilon} \frac{2m}{\hbar^2} E \psi dx - \int_{-\epsilon}^{\epsilon} \frac{2m}{\hbar^2} V_0 a \delta(x) \psi dx \end{aligned}$$

$$= 0 - \frac{2m}{\hbar^2} V_0 a \psi(x) \Big|_{-\epsilon}^{\epsilon} \quad (2)$$

$\psi(x)$ is continuous δ -function integral

Take the limit $\epsilon \rightarrow 0$

$$(1) \rightarrow \lim_{\epsilon \rightarrow 0} \psi'' dx = \psi'_I(0) - \psi'_I(0)$$

$$(2) \rightarrow \lim_{\epsilon \rightarrow 0} \psi'' dx = - \frac{2m}{\hbar^2} V_0 a \psi(0)$$

Therefore, the boundary conditions for δ -potential are:

$$\begin{cases} \psi_I(0) = \psi_{II}(0) \end{cases} \quad (3)$$

$$\begin{cases} \psi'_I(0) - \psi'_{II}(0) = - \frac{2m}{\hbar^2} V_0 a \psi(0) \end{cases} \quad (4)$$

Let's solve the problem 5.2.3.

$$\text{I. } \psi'' + \frac{2m}{\hbar^2} E \psi = 0, \quad \text{set } -\frac{2m}{\hbar^2} E = k^2 \quad (E < 0)$$

$$\Rightarrow \psi = A e^{kx} + B e^{-kx}$$

Since $\psi \rightarrow 0$ as $x \rightarrow -\infty$, B should be 0.

$$\text{Therefore, } \psi_{\text{I}}(x) = A e^{kx} \quad (5)$$

$$\text{II. } \psi'' + \frac{2m}{\hbar^2} E \psi = 0, \quad \text{set } -\frac{2m}{\hbar^2} E = k^2 \quad (E < 0)$$

$$\Rightarrow \psi = C e^{kx} + D e^{-kx}$$

Since $\psi \rightarrow 0$ as $x \rightarrow \infty$, C should be 0.

$$\text{Therefore, } \psi_{\text{II}}(x) = D e^{-kx} \quad (6)$$

Substitute (5) and (6) into the boundary conditions (3) and (4), we have,

$$\begin{cases} D = A \\ -kD - kA = -\frac{2m}{\hbar^2} V_0 a D \end{cases}$$

$$\Rightarrow k = \frac{m}{\hbar^2} V_0 a$$

$$\Rightarrow E = -\frac{\hbar^2}{2m} k^2 = -\frac{m a^2 V_0^2}{2\hbar^2}$$

This is the only solution for E . (Only one bound state.)

5.2.4

The energy of a particle in a box is given by

$$E_n = \frac{\hbar^2 \pi^2 n^2}{2mL^2}$$

$$F = -\frac{\partial E_n}{\partial L} = \frac{\hbar^2 \pi^2 n^2}{mL^3} = \frac{2E_n}{L}$$

From the classical point of view, when a particle hits the wall,

$$\Delta P = P - (-P) = 2P \quad (\text{change of the momentum})$$

The average time interval between the particle hits the same wall twice:

$$\overline{\Delta t} = \frac{2L}{v}$$

Considering $E = \frac{P^2}{2m} = \frac{1}{2} m v^2$, and $\overline{F} = \frac{\Delta P}{\Delta t}$, we have

$$\overline{F} = \frac{PV}{L} = \frac{2E}{L}$$

This agrees with the quantum result above.

5.2.5

Between 0 and L, we have:

$$\psi'' + \frac{2m}{\hbar^2} E \psi = 0.$$

Set $k^2 = \frac{2m}{\hbar^2} E$,

$$\psi'' + k^2 \psi = 0,$$

$$\psi(x) = A \sin(kx) + B \cos(kx).$$

Boundary conditions,

$$\psi(0) = 0 \Rightarrow B = 0$$

$$\psi(L) = 0 \Rightarrow A \sin(kL) = 0 \Rightarrow kL = n\pi \Rightarrow k = \frac{n\pi}{L}, \quad n = 1, 2, 3, \dots$$

Therefore,

$$E_n = \frac{\hbar^2 k^2}{2m} = \frac{n^2 \hbar^2 \pi^2}{2mL^2}.$$

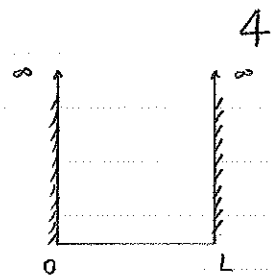
$$\psi(x) = A \sin\left(\frac{n\pi x}{L}\right).$$

Normalization,

$$1 = \int_0^L \psi^*(x) \psi(x) dx = A^2 \int_0^L \sin^2\left(\frac{n\pi x}{L}\right) dx = A^2 \cdot \frac{L}{2}$$

$$\Rightarrow A = \sqrt{\frac{2}{L}}.$$

$$\Rightarrow \psi(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right).$$



5.2.6

(i) II: $\psi'' + \frac{2m}{\hbar^2} E \psi = 0$

$$k^2 = \frac{2m}{\hbar^2} E,$$

$$\Rightarrow \psi_{II}(x) = A \cos kx + B \sin kx$$

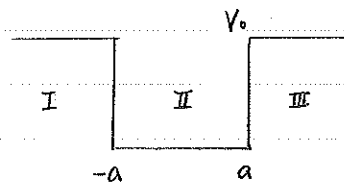
I and III: $\psi'' + \frac{2m}{\hbar^2} (E - V_0) \psi = 0$

$$k^2 = -\frac{2m}{\hbar^2} (E - V_0)$$

$$\Rightarrow \psi(x) = C e^{kx} + D e^{-kx}$$

Since $\psi_I(x) \rightarrow 0$ as $x \rightarrow -\infty$; $\psi_{III}(x) \rightarrow 0$ as $x \rightarrow \infty$, we have

$$\psi_I(x) = C e^{kx}, \quad \psi_{III}(x) = D e^{-kx}.$$



* For even solution:

$$\begin{cases} \psi_{II}(x) = A \cos kx \\ \psi_{III}(x) = D e^{-kx} \end{cases}$$

Boundary conditions,

$$\begin{cases} \psi_{II}(a) = \psi_{III}(a) \\ \psi'_{II}(a) = \psi'_{III}(a) \end{cases} \Rightarrow \begin{cases} A \cos ka = D e^{-ka} \\ -kA \sin ka = -kD e^{-ka} \end{cases}$$

$$\Rightarrow k \tan ka = k.$$

* For odd solution:

$$\begin{cases} \psi_{II}(x) = B \sin kx \\ \psi_{III}(x) = D e^{-kx} \end{cases}$$

Boundary conditions,

$$\begin{cases} \psi_{II}(a) = \psi_{III}(a) \\ \psi'_{II}(a) = \psi'_{III}(a) \end{cases} \Rightarrow \begin{cases} B \sin ka = D e^{-ka} \\ kB \cos ka = -kD e^{-ka} \end{cases}$$

$$\Rightarrow k \cot ka = -k.$$

$$k^2 + k^2 = \frac{2m}{\hbar^2} E - \frac{2m}{\hbar^2} (E - V_0) = \frac{2m}{\hbar^2} V_0$$

* When $V_0 \rightarrow \infty$, $k \rightarrow \infty$ while α keeps the same.

Therefore, $\tan ka$, $\cot ka \rightarrow \infty$

$$ka = \frac{n\pi}{2}, \quad n = 1, 2, 3, \dots$$

Thus, $E = \frac{\hbar^2}{2m} k^2 = \frac{\hbar^2}{2m} \cdot \frac{n^2 \pi^2}{4a^2} = \frac{n^2 \pi^2 \hbar^2}{8ma^2}$. This is the energy level in the box with length $2a$.

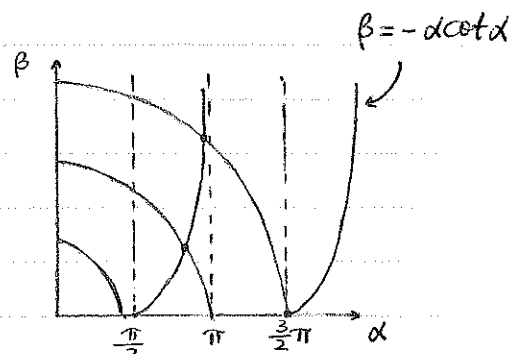
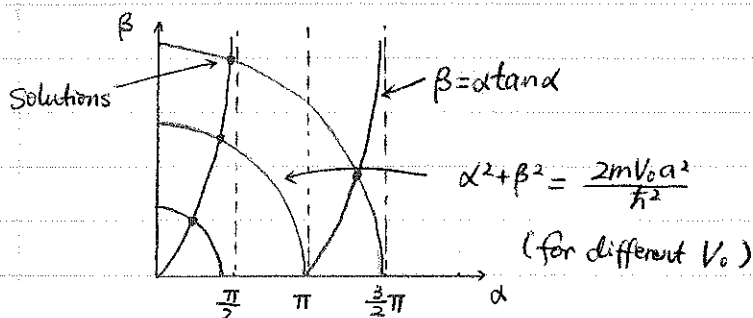
(2)

$$\beta = ka, \quad \alpha = ka$$

$$\Rightarrow \beta = \alpha \tan \alpha \quad \text{for even solution,}$$

$$\Rightarrow \beta = -\alpha \cot \alpha \quad \text{for odd solution.}$$

$$\alpha^2 + \beta^2 = a^2 (k^2 + k^2) = \frac{2mV_0 a^2}{\hbar^2}.$$



- (3) From part (2), we see that there always exists at least one intersection between $\alpha^2 + \beta^2 = \frac{2mV_0 a^2}{\hbar^2}$ and $\beta = \alpha \tan \alpha$, no matter how small V_0 is. Therefore, there is always one even solution. This accords with Exercise 5.2.2 b.

However, we cannot find an intersection between $\alpha^2 + \beta^2 = \frac{2mV_0 a^2}{\hbar^2}$ and $\beta = -\alpha \cot \alpha$ unless $\sqrt{\frac{2mV_0 a^2}{\hbar^2}} \geq \frac{\pi}{2}$. Therefore, there is no odd solution unless $V_0 \geq \hbar^2 \pi^2 / 8ma^2$. When V_0 just meets this requirement, we have the intersection $(\alpha, \beta) = (\frac{\pi}{2}, 0)$. Hence,

$$0 = \beta = ka = \sqrt{-\frac{2m}{\hbar^2}(E - V_0)} \cdot a$$

$$\Rightarrow E = V_0 = -\frac{\hbar^2 \pi^2}{8ma^2}.$$

Problem 2

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In[3]:= Psi[x_, t_] := (Pi (Δ^2 + hbar^2 t^2 / (m^2 Δ^2))) ^ (-1 / 2)
      Exp[(- (x - (p0 / m) t) ^ 2) / (Δ^2 + hbar^2 t^2 / (m^2 Δ^2))];
Plot[Evaluate[Table[Psi[x, t] /. {p0 → 1, Δ → 1, hbar → 1, m → 1},
      {t, 0, 14}]], {x, -20, 50},
      AxesLabel → {"x", "|ψ(x)|^2"}, PlotRange → All]

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Out[4]=

